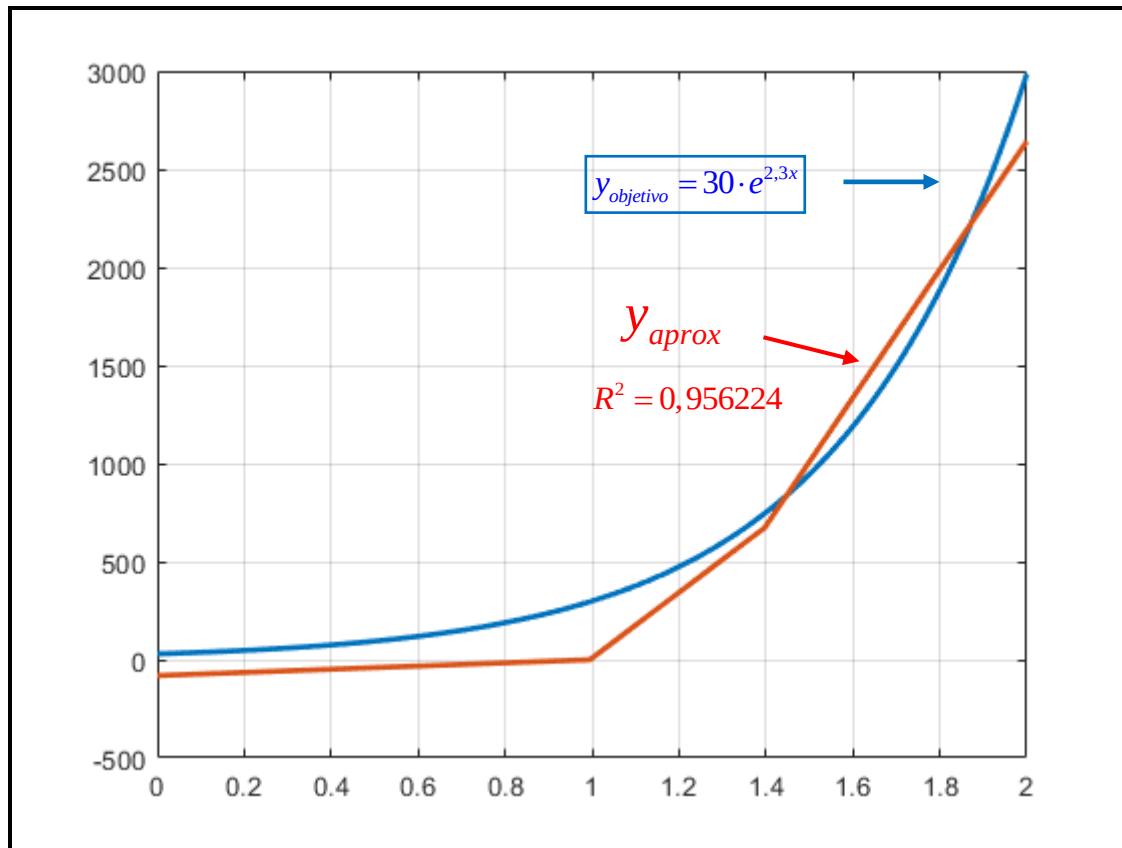
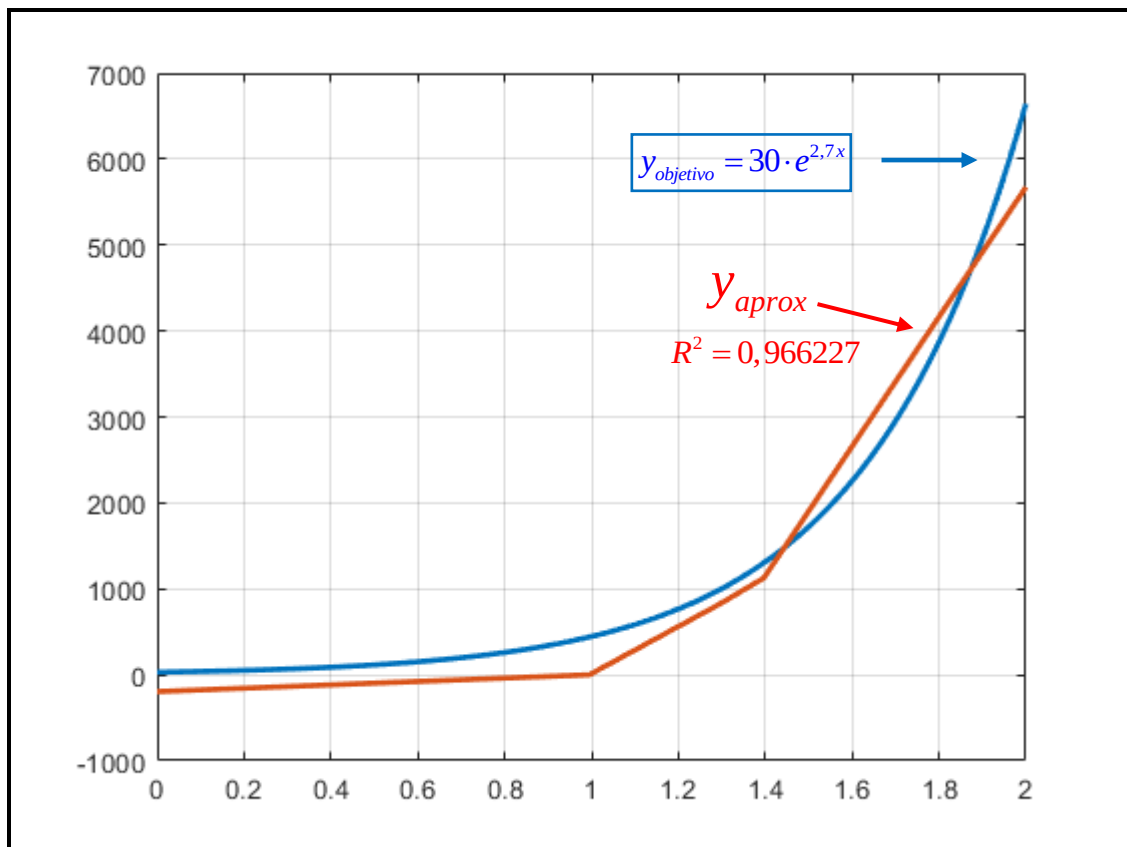


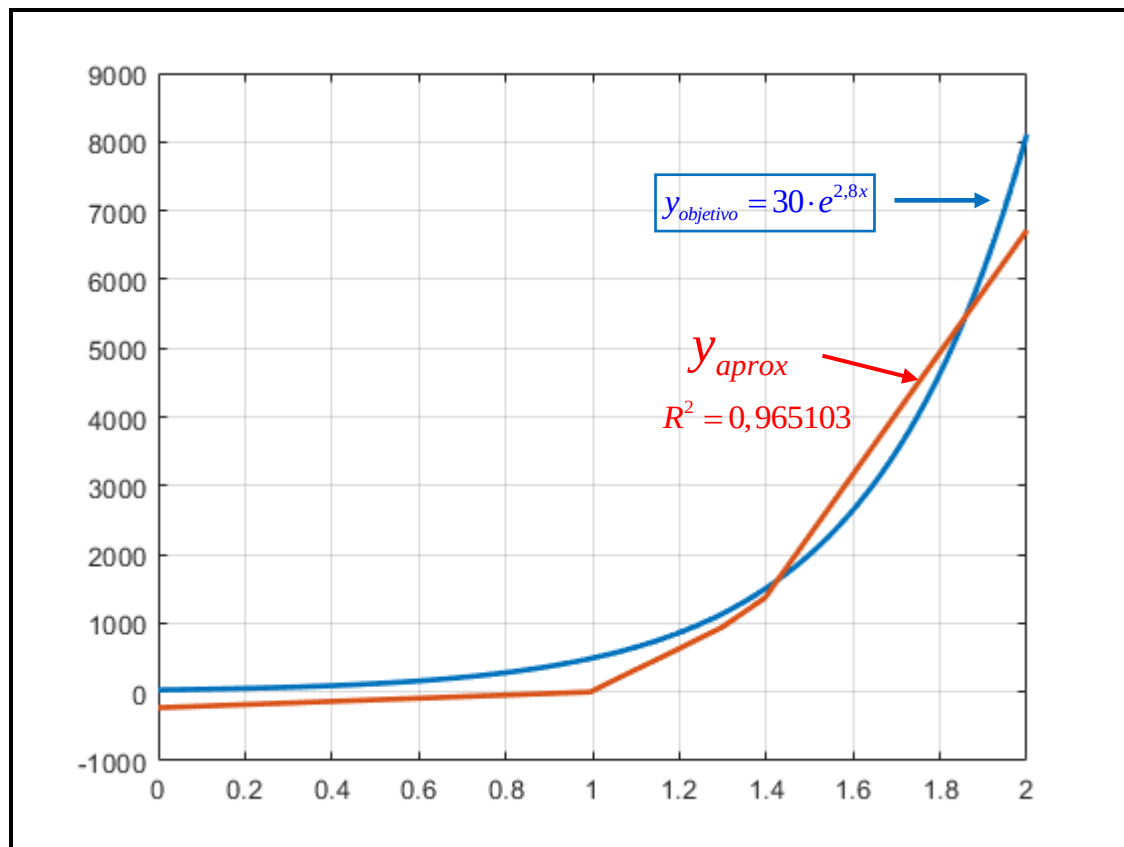
Aproximación de la función objetivo con una combinación lineal de coeficientes enteros para 5 funciones base, realizado con una Red Neuronal entrenada. Los datos de entrenamiento se logran buscando los valores mas cercanos a los óptimos con el método de los Multiplicadores de Lagrange (reales) + MIQP (para redondeo a enteros)



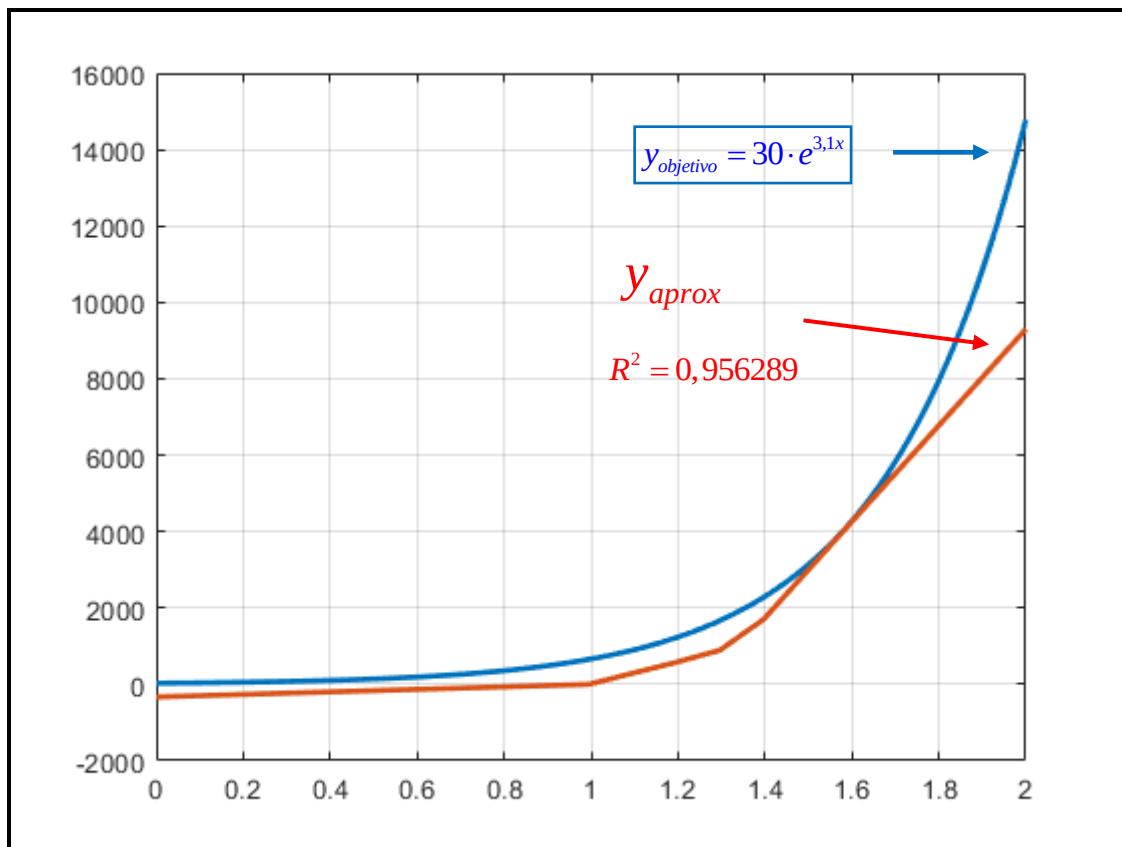
$$y_{\text{aprox}} = 5 \cdot \text{base}_1 + 0 \cdot \text{base}_2 + 0 \cdot \text{base}_3 + 0 \cdot \text{base}_4 + 7 \cdot \text{base}_5$$



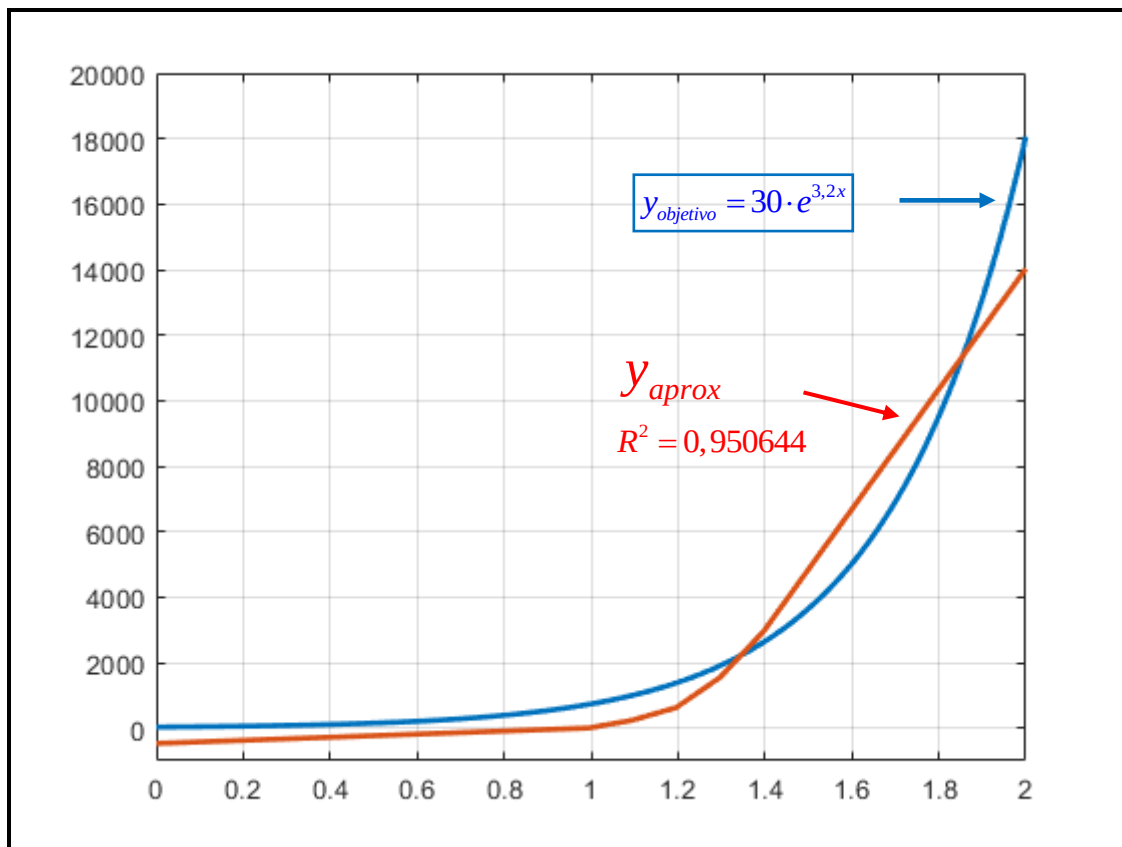
$$y_{\text{aprox}} = 8 \cdot \text{base}_1 + 0 \cdot \text{base}_2 + 0 \cdot \text{base}_3 + 1 \cdot \text{base}_4 + 20 \cdot \text{base}_5$$



$$y_{aprox} = 9 \cdot base_1 + 0 \cdot base_2 + 0 \cdot base_3 + 5 \cdot base_4 + 20 \cdot base_5$$

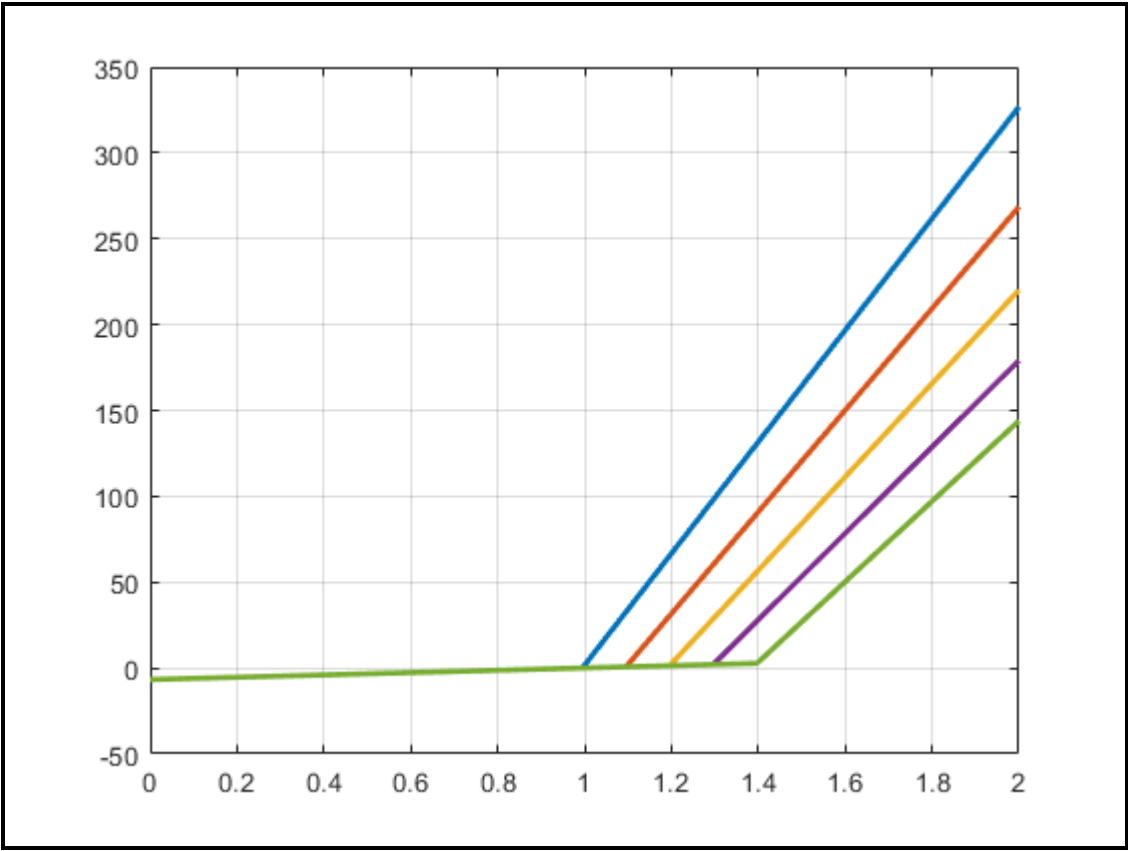


$$y_{\text{aprox}} = 8 \cdot \text{base}_1 + 0 \cdot \text{base}_2 + 12 \cdot \text{base}_3 + 20 \cdot \text{base}_4 + 20 \cdot \text{base}_5$$



$$y_{\text{aprox}} = 6 \cdot \text{base}_1 + 4 \cdot \text{base}_2 + 20 \cdot \text{base}_3 + 20 \cdot \text{base}_4 + 20 \cdot \text{base}_5$$

Funciones Base



Rango de variacion de las funciones objetivo para aproximaciones con $R^2 > 0,94$

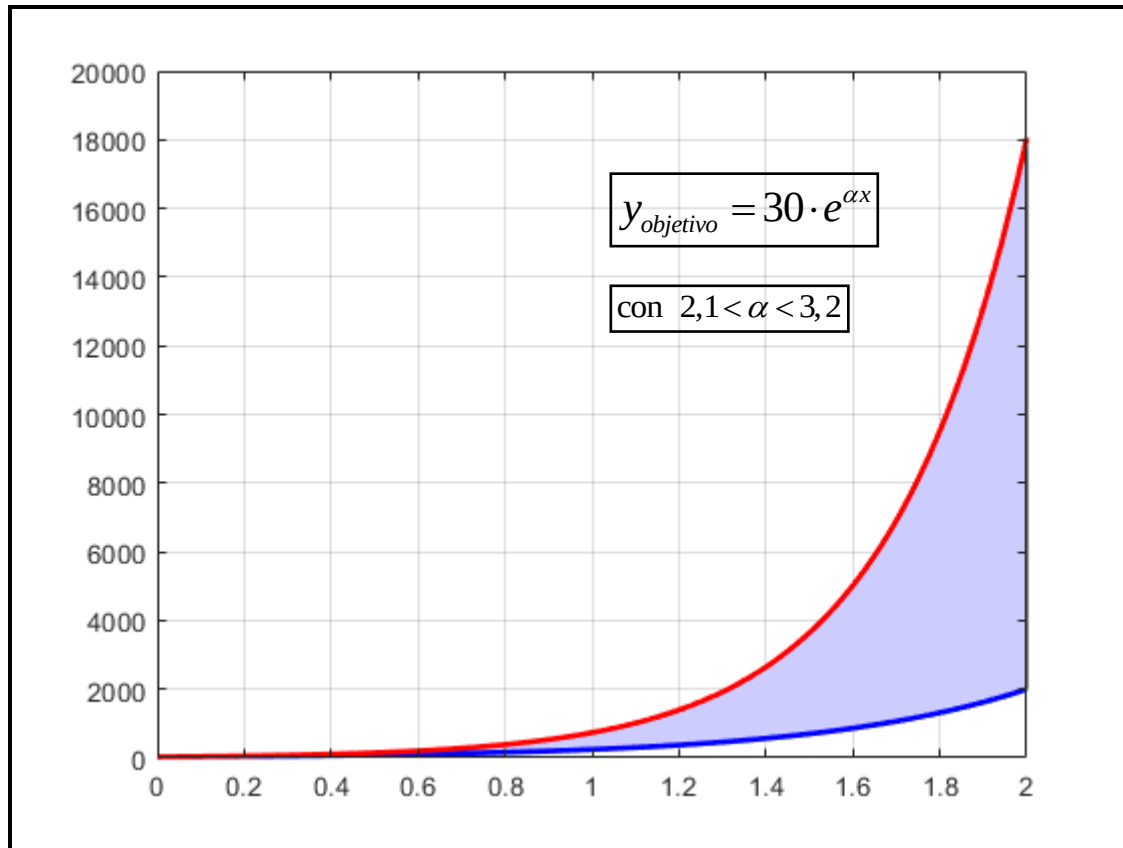


TABLA DE REFERENCIA (Multiplicadores de Lagrange + MIQP)

α	Coeficientes	R^2
2.0	[4, 0, 0, 0, 1]	0.933719
2.1	[4, 0, 0, 0, 3]	0.942820
2.2	[4, 1, 0, 0, 4]	0.949399
2.3	[5, 0, 0, 0, 7]	0.956224
2.4	[6, 0, 0, 0, 8]	0.960104
2.5	[7, 0, 0, 0, 13]	0.962868
2.6	[7, 0, 0, 0, 17]	0.965206
2.7	[8, 0, 0, 1, 20]	0.966227
2.8	[9, 0, 0, 5, 20]	0.965103
2.9	[9, 0, 0, 13, 20]	0.963391
3.0	[6, 0, 11, 15, 19]	0.959045
3.1	[8, 0, 12, 20, 20]	0.956289
3.2	[6, 4, 20, 20, 20]	0.950644